



CCDSALG Project 2 Specifications
Term 3, AY 2025–2026
Deadline: **Aug 3, 2026 (M)** before 0800

1 Overview

This project models the historic *rat in a maze* psychology experiment as a computational pathfinding problem. Using C or Java, you will create a TUI- or GUI-based grid simulation where a character must find its way from a starting cell to a goal cell. Your program must implement a complete search algorithm to guarantee a solution is found whenever a valid path exists. Your tasks include:

Algorithm. Implement a Graph Traversal Algorithm for maze navigation.

Data Structures. Use self-implemented Stacks, Queues, or Priority Queues to handle backtracking and exploration states.

UI. Build a text-based or graphical interface to visualize the entire pathfinding process.

2 Application Requirements

2.1 The Maze

The maze is represented as an $M \times N$ 2D-grid, where $15 \leq M \leq 30, 15 \leq N \leq 30$. Each cell can either be an open path, or a wall. One of the open paths may be the start position of the character, and another one may be the goal position.

2.2 Movement

The character can move up, down, left, or right onto any open path or the goal. Diagonal movements are not allowed.

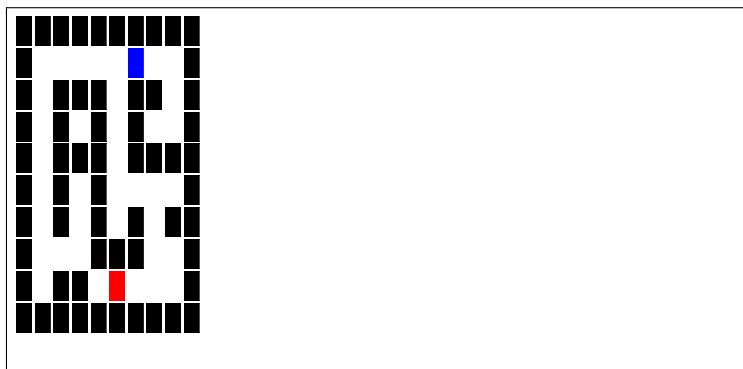
2.3 Maze File

Load the maze from a text file where the first two lines define the dimensions (M and N , respectively). The remaining lines contain the maze grid, using the following symbols:

Symbol	Description
<i>(space)</i>	open path
#	wall
S	rat's starting point
G	location of cheese (goal)

The following is a sample maze file (on the left), and the corresponding sampl maze visualization (on the right).

```
10 10
#####
#      S  #
#  ##  ## #
# # # # #
#  ##  ####
# # #   #
# # # # ##
#   ##  #
# ## G  #
#####
```



3 Interface Requirements

Your application should be **menu-driven**, with the following elements:

- **Menu.** Include options to Load Maze, Start Simulation, and Exit.
- **Display.** Show the maze visually with the character, the walls, open paths, and the goal.

- **Animation.** Animate the pathfinding process in real-time (e.g., using a timer delay) so the user can see the algorithm exploring paths, hitting dead ends, and backtracking. Do not just instantly warp the character to the finish line.
- **Metrics Display.** Show live statistics once the search concludes:
 - Total steps taken/cells explored.
 - Final path length.
 - Execution time (in milliseconds).

4 Implementation

Apply good programming practices by writing well-structured, modular code with the use of clear and descriptive variable names.

The following rules will be strictly enforced. Each violation will lead to a deduction from your project score.

- declaration and use of global variables is not allowed;
- **break** statements may only be found inside **switch** statements;
- **return** statements may only be found in functions where the return type is not **void**;
- **return** statements may not be used to prematurely terminate a loop, a function, or a program;
- the use of a **continue** statement is not allowed;
- the use of **exit()** statement is not allowed;
- the use of **goto** labels is not allowed; and
- calling the **main()** function is not allowed.

5 Deliverables

The following must be submitted via **AnimoSpace** on or before **Aug 3, 2026 (M), 0800**.

1. Poster

Upload a one-page A4 academic poster detailing your system's architecture and findings. The poster must clearly present the following:

- Introduction**
- Data Structure Design.** Discusses how the maze and state of the character are represented. Discuss also other data structures used in your solution (e.g. Stacks, Queues, Priority Queues, etc.).
- Algorithm Design and Theoretical Analysis.** Discuss your solution. Include a diagram or pseudocode to support the description. Provide the Big-Oh for different cases. Include space complexity (memory usage) for worst case scenario.
- Discussion.** Discuss how different maze types (no solution, single and multiple solution) are handled by your algorithm.
- Experiment Results.** Describe your test environment (HW specifications), metrics and comparison of execution time of different maze dimensions and types.
- Conclusion and Recommendation**
- References.** List of references for algorithms and literature used in APA format.

2. Source Code and Datasets

Upload the following:

- Source code files (**.h**, **.c**, or **.java**)
- Maze files (**.txt**)
- Zip file containing source codes, datasets, etc. and a **README** file with the exact terminal commands required to compile and run your application. (**.zip**)

3. Declarations

Upload a PDF file with the following.

(a) AI Disclosure

State whether AI tools were used as a *learning partner* during this project. Directly copying AI-generated code is **strictly forbidden**. If AI was used, you must document the specific tools, the extent of their use, your exact prompts, and a breakdown of what you modified versus what was used as-is. Links or transcripts of the full conversation threads must be attached in the Appendix at the end of this document.

(b) Contribution Declaration

List the individual contributions of each group member to the project.

6 Rubric for Grading

Table 1: Rubrics for CCDSALG MCO2, Term 3, AY 2025-2026

CRITERIA	COMPLETE	INCOMPLETE	NO MARKS
DATA REPRESENTATION (20 points)	20 points Proper choice of data representation and correct implementation of maze, character state and supporting data structures.	10 points Choice of data representation or implementation is not entirely correct.	0 point Not implemented.
ALGORITHM IMPLEMENTATION (30 points)	30 points Correctly implemented the pathfinding algorithm with proper use of supporting data structures.	15 points Implementation is not entirely correct.	0 point Not implemented.
USER INTERFACE (25 points)	25 points Menu-driven, Interface is neat and organize. Search process is animated and metrics are displayed.	12 points Implementation is not entirely correct. Some elements are missing or incomplete.	0 point Not implemented.
POSTER (15 points)	15 points Required details were covered and clearly presented. Appropriate visualizations are used.	8 points Some required details were not discussed.	0 point No discussion or documentation.
DEMO AND Q&A (10 points)	10 points Group members can confidently answer all questions during the demo.	5 points Group members can answer all questions during the demo, with some unconvincing or illogical responses.	0 point The group failed to answer most questions during the demo.